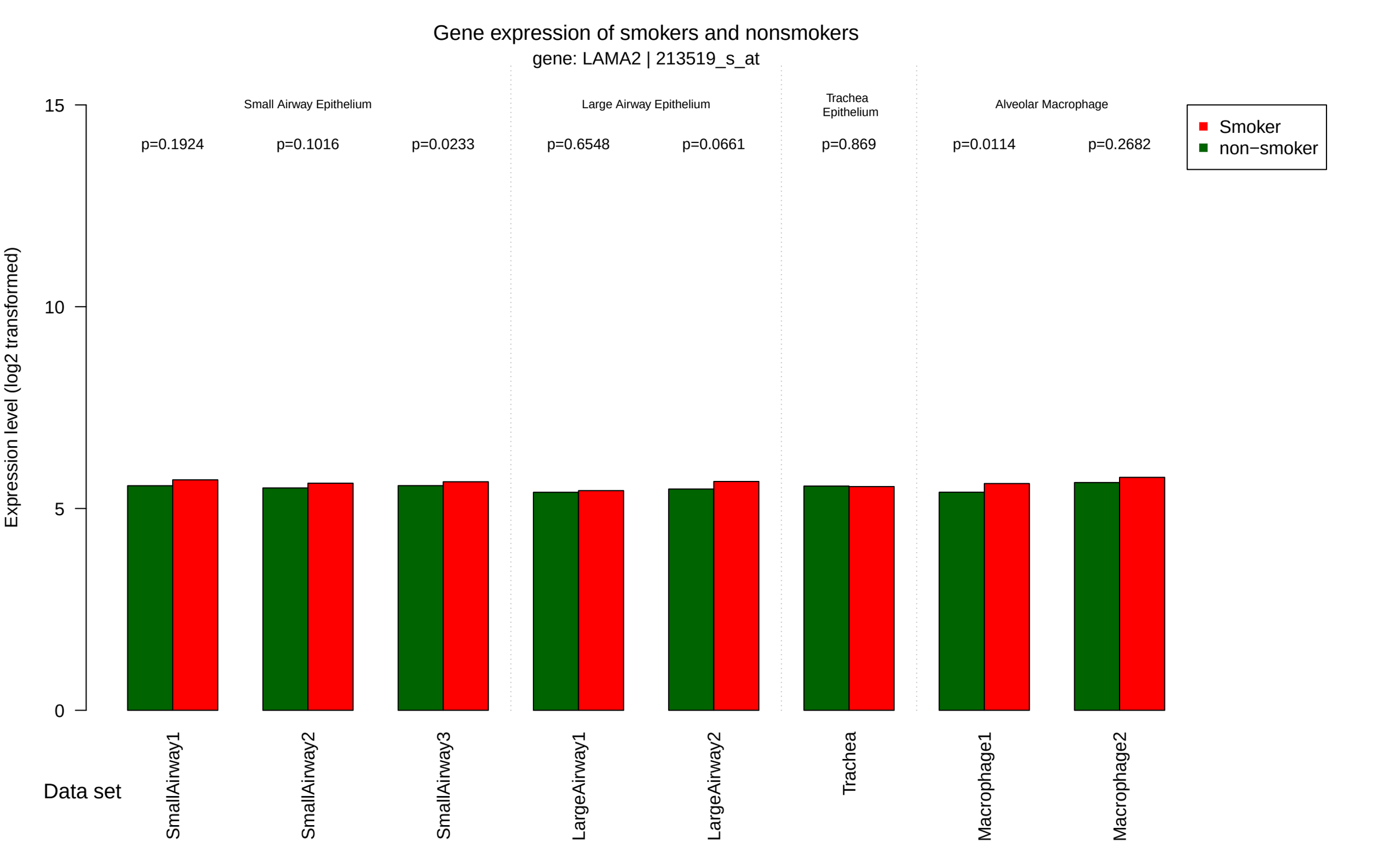
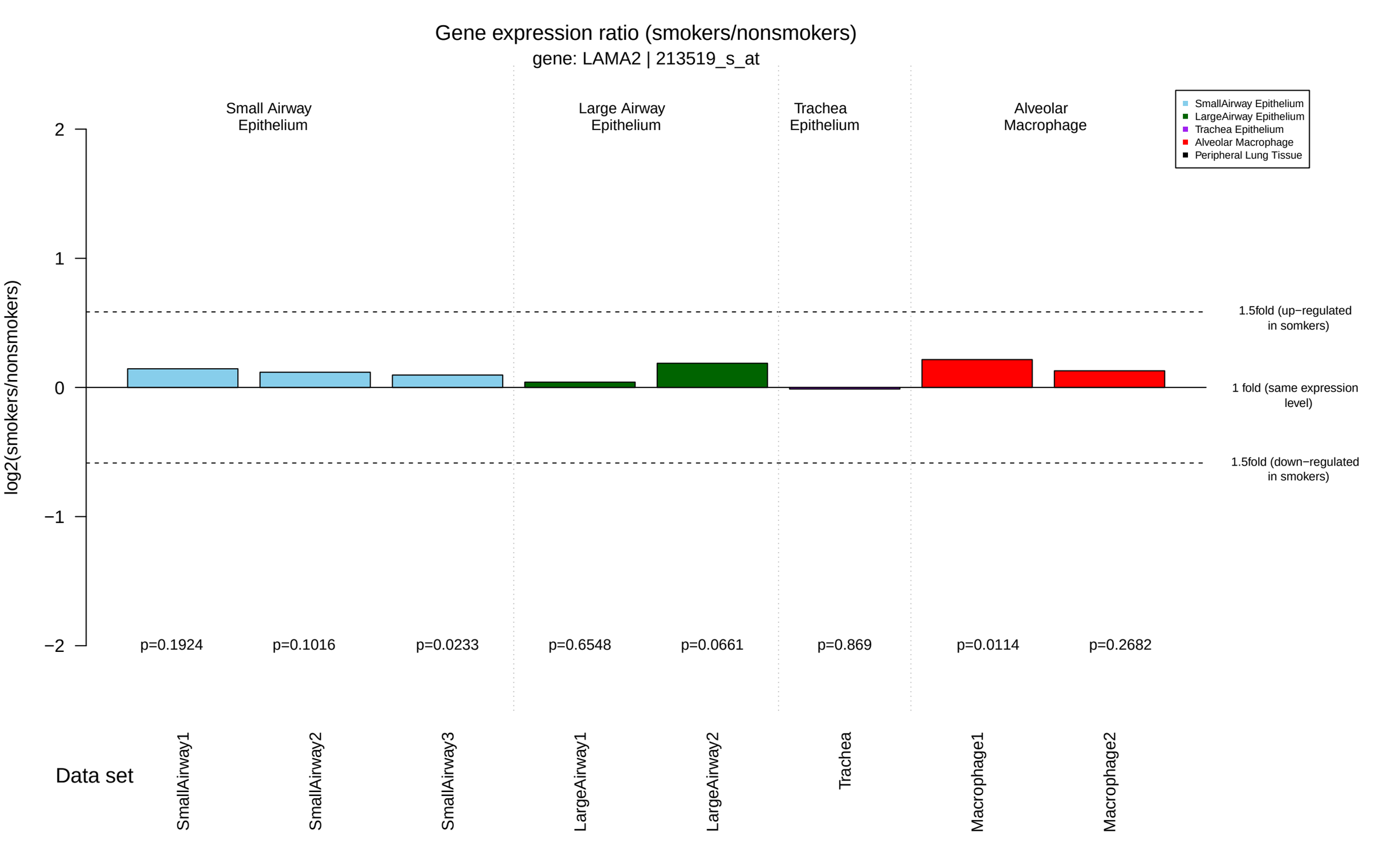
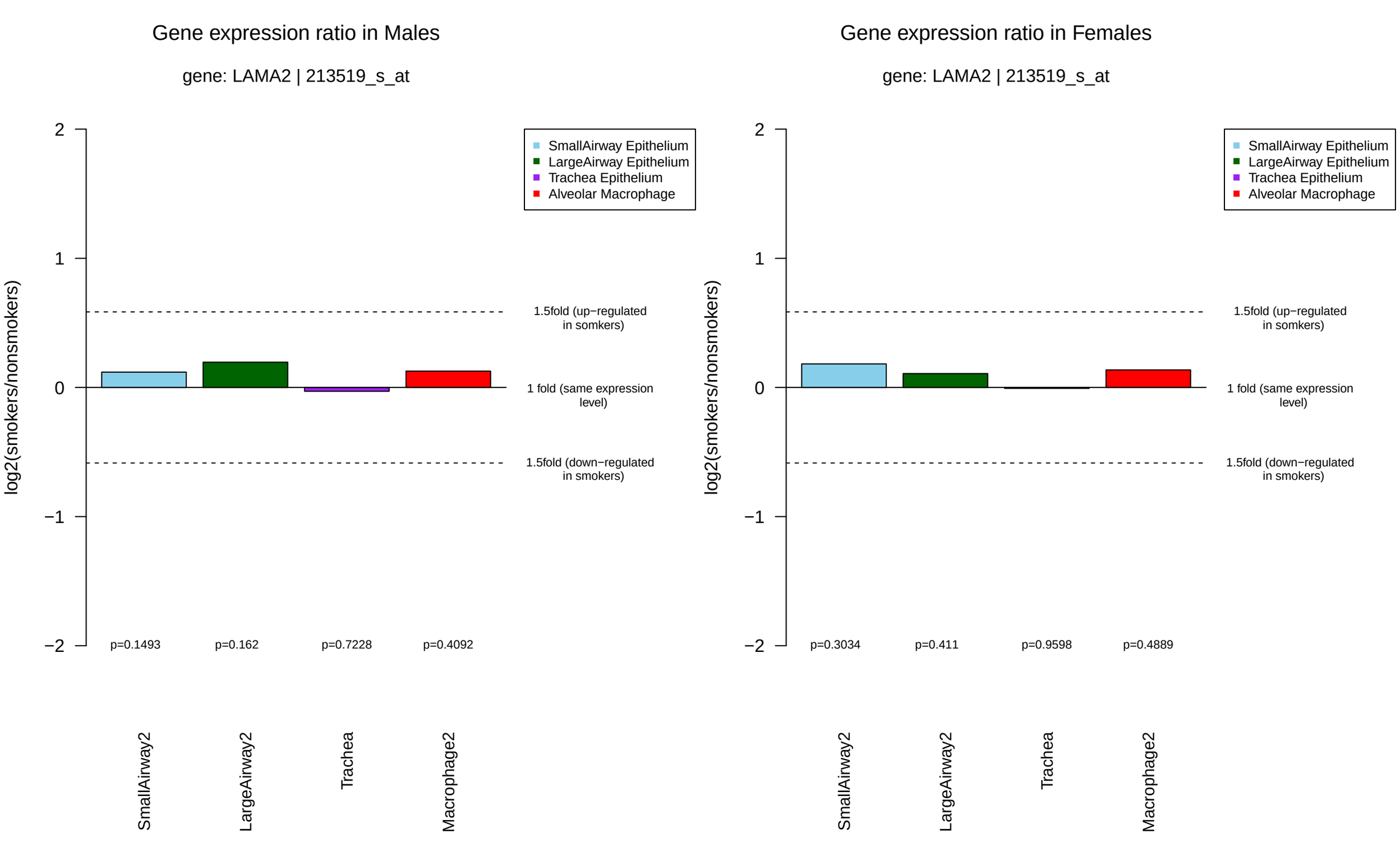
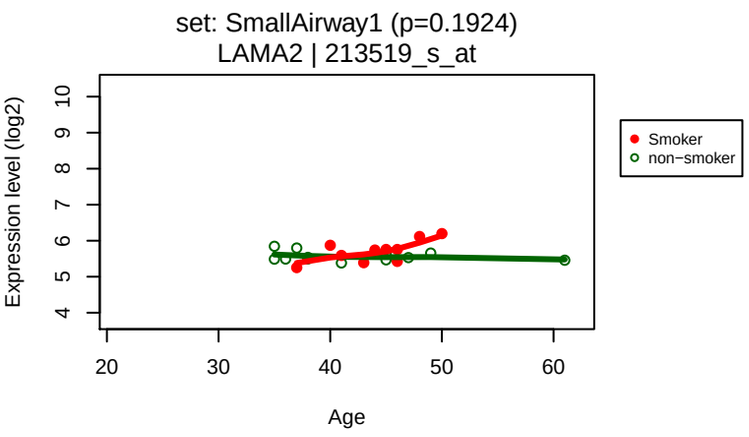
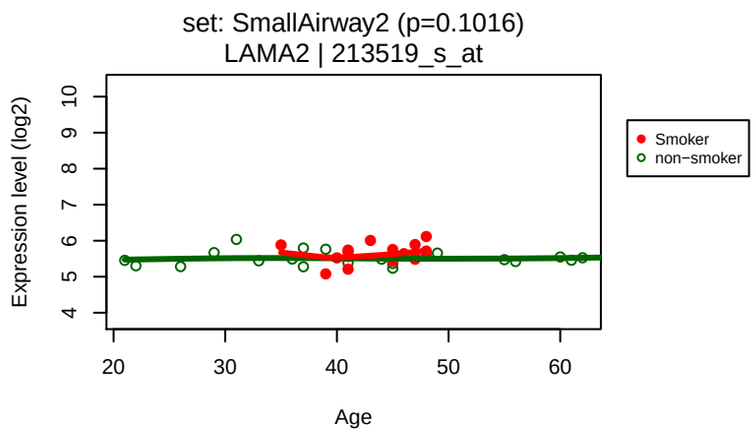
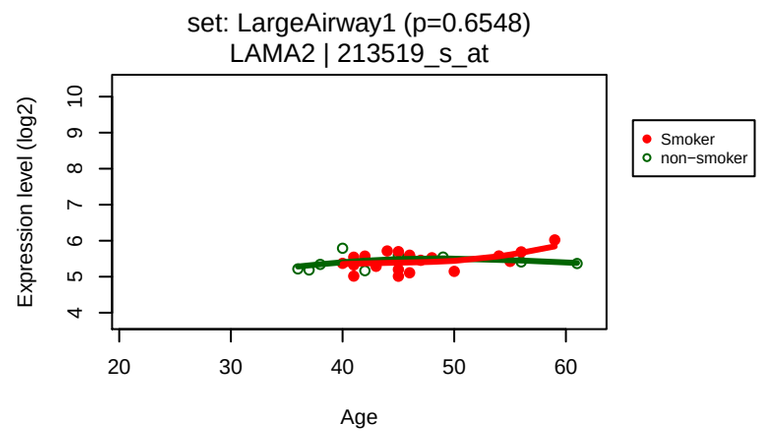

Gene expression of smokers and nonsmokers
gene: LAMA2 | 213519_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1924
p=0.1016
p=0.0233
p=0.6548
p=0.0661
p=0.869
p=0.0114
p=0.2682
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

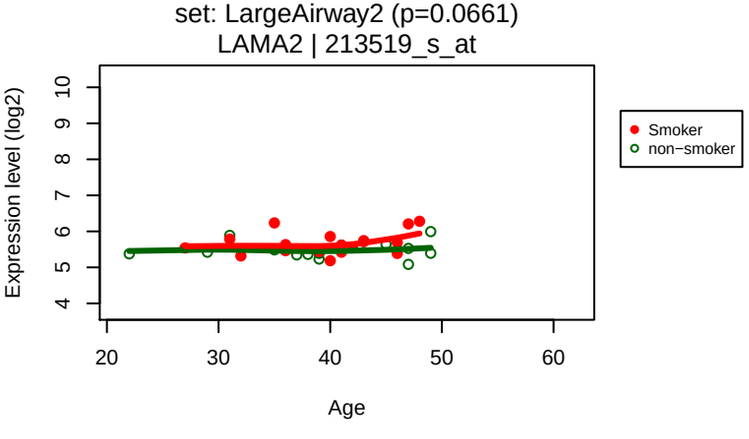

Gene expression ratio (smokers/nonsmokers)
gene: LAMA2 | 213519_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1924
p=0.1016
p=0.0233
p=0.6548
p=0.0661
p=0.869
p=0.0114
p=0.2682
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

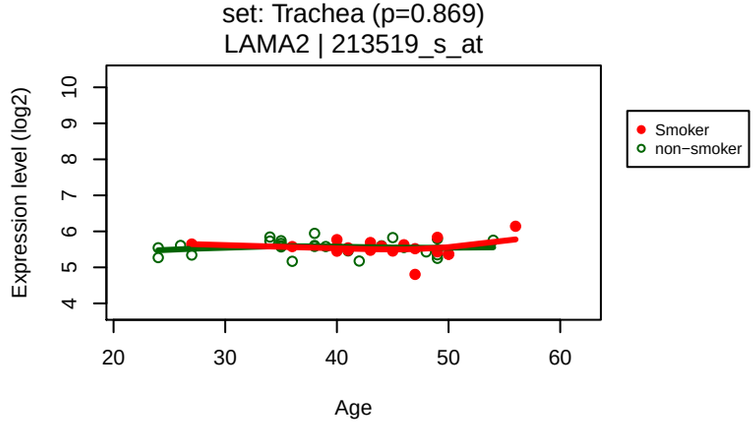

Gene expression ratio in Males
gene: LAMA2 | 213519_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1493
p=0.162
p=0.7228
p=0.4092
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LAMA2 | 213519_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3034
p=0.411
p=0.9598
p=0.4889
SmallAirway2
LargeAirway2
Trachea
Macrophage2

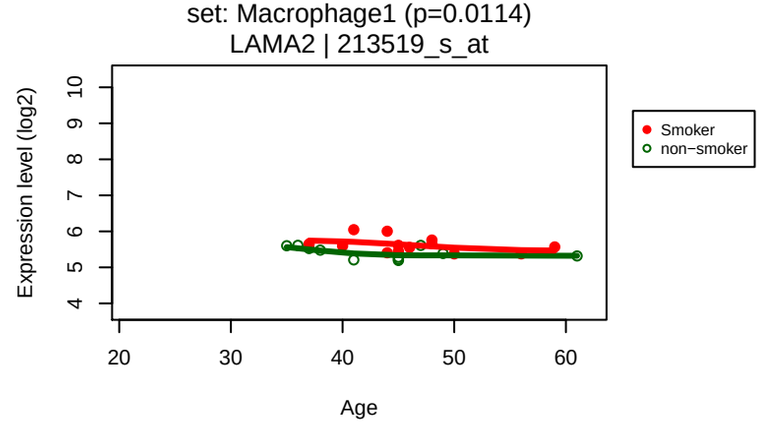

set: SmallAirway1 (p=0.1924)
LAMA2 | 213519_s_at
Expression level (log2)
Age
Smoker
non−smoker

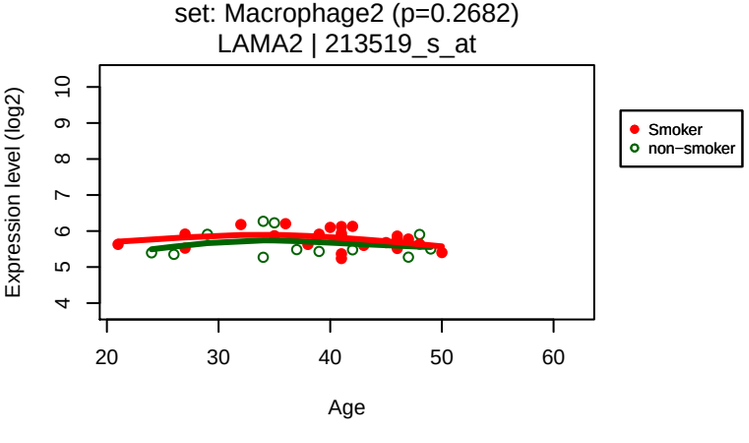

set: SmallAirway2 (p=0.1016)
LAMA2 | 213519_s_at
Expression level (log2)
Age
Smoker
non−smoker

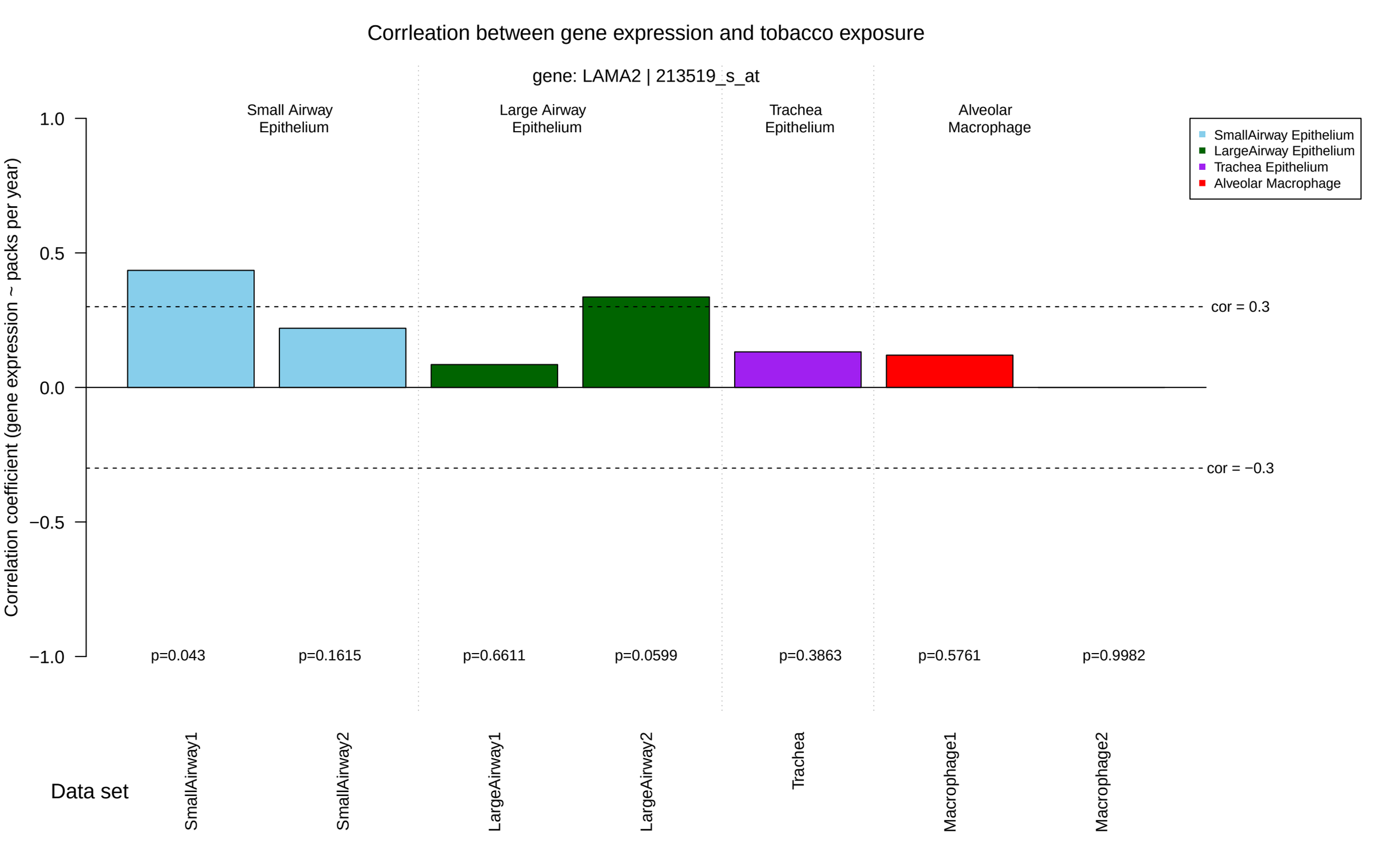

set: LargeAirway1 (p=0.6548)
LAMA2 | 213519_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0661)
LAMA2 | 213519_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.869)
LAMA2 | 213519_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0114)
LAMA2 | 213519_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2682)
LAMA2 | 213519_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LAMA2 | 213519_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.043
p=0.1615
p=0.6611
p=0.0599
p=0.3863
p=0.5761
p=0.9982
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2